

Results of calculations for numbers in IEEE 754 standard and NPAAt format

Comparative results of summing signed decimal numbers according to the standard IEEE 754 algorithm and summation in NPAAt format.

The terminal screenshots show the output data obtained using the demonstration algorithm `npap-algorithm.cpp`

Sum of $x_1 = -0.893827$ with 999 additions of
 $x_2 = 0.787827$

```
x1 : -0.893827
x2 : 0.787827
Z : 1000
```

i= 2		NPAp =	-0.106		IEEE =	-0.106		Match: YES
i= 3		NPAp =	0.681827		IEEE =	0.681827		Match: YES
i= 4		NPAp =	1.46965		IEEE =	1.46965		Match: YES
i= 5		NPAp =	2.25748		IEEE =	2.25748		Match: YES
i= 6		NPAp =	3.04531		IEEE =	3.04531		Match: YES
i= 7		NPAp =	3.83314		IEEE =	3.83314		Match: YES
i= 8		NPAp =	4.62096		IEEE =	4.62096		Match: YES
i= 9		NPAp =	5.40879		IEEE =	5.40879		Match: YES
i= 10		NPAp =	6.19662		IEEE =	6.19662		Match: YES
i=100		NPAp =	77.1011		IEEE =	77.1011		Match: YES
i=200		NPAp =	155.884		IEEE =	155.884		Match: YES
i=300		NPAp =	234.666		IEEE =	234.666		Match: YES
i=400		NPAp =	313.449		IEEE =	313.449		Match: YES
i=500		NPAp =	392.232		IEEE =	392.232		Match: YES
i=600		NPAp =	471.015		IEEE =	471.015		Match: YES
i=700		NPAp =	549.797		IEEE =	549.797		Match: YES
i=800		NPAp =	628.58		IEEE =	628.58		Match: YES
i=900		NPAp =	707.363		IEEE =	707.363		Match: YES
i=1000		NPAp =	786.146		IEEE =	786.146		Match: YES

2. Sum of $x_1 = -0.893827$ with 9999 additions
of $x_2 = 0.787827$

```
x1 : -0.893827
x2 : 0.787827
Z : 10000
```

i= 2		NPAp =	-0.106		IEEE =	-0.106		Match: YES
i= 3		NPAp =	0.681827		IEEE =	0.681827		Match: YES
i= 4		NPAp =	1.46965		IEEE =	1.46965		Match: YES
i= 5		NPAp =	2.25748		IEEE =	2.25748		Match: YES
i= 6		NPAp =	3.04531		IEEE =	3.04531		Match: YES
i= 7		NPAp =	3.83314		IEEE =	3.83314		Match: YES
i= 8		NPAp =	4.62096		IEEE =	4.62096		Match: YES
i= 9		NPAp =	5.40879		IEEE =	5.40879		Match: YES
i= 10		NPAp =	6.19662		IEEE =	6.19662		Match: YES
i=1000		NPAp =	786.146		IEEE =	786.146		Match: YES
i=2000		NPAp =	1573.97		IEEE =	1573.97		Match: YES
i=3000		NPAp =	2361.8		IEEE =	2361.8		Match: YES
i=4000		NPAp =	3149.63		IEEE =	3149.63		Match: YES
i=5000		NPAp =	3937.45		IEEE =	3937.45		Match: YES
i=6000		NPAp =	4725.28		IEEE =	4725.28		Match: YES
i=7000		NPAp =	5513.11		IEEE =	5513.11		Match: YES
i=8000		NPAp =	6300.94		IEEE =	6300.94		Match: YES
i=9000		NPAp =	7088.76		IEEE =	7088.76		Match: YES

3. Sum of $x_1 = -0.893827$ with 99999
additions of $x_2 = 0.787827$

```
x1 : -0.893827
x2 : 0.787827
Z : 100000
```

i= 2		NPAp =	-0.106		IEEE =	-0.106		Match: YES
i= 3		NPAp =	0.681827		IEEE =	0.681827		Match: YES
i= 4		NPAp =	1.46965		IEEE =	1.46965		Match: YES
i= 5		NPAp =	2.25748		IEEE =	2.25748		Match: YES
i= 6		NPAp =	3.04531		IEEE =	3.04531		Match: YES
i= 7		NPAp =	3.83314		IEEE =	3.83314		Match: YES
i= 8		NPAp =	4.62096		IEEE =	4.62096		Match: YES
i= 9		NPAp =	5.40879		IEEE =	5.40879		Match: YES
i= 10		NPAp =	6.19662		IEEE =	6.19662		Match: YES
i=10000		NPAp =	7876.59		IEEE =	7876.59		Match: YES
i=20000		NPAp =	15754.9		IEEE =	15754.9		Match: YES
i=30000		NPAp =	23633.1		IEEE =	23633.1		Match: YES
i=40000		NPAp =	31511.4		IEEE =	31511.4		Match: YES
i=50000		NPAp =	39389.7		IEEE =	39389.7		Match: YES
i=60000		NPAp =	47267.9		IEEE =	47267.9		Match: YES
i=70000		NPAp =	55146.2		IEEE =	55146.2		Match: YES
i=80000		NPAp =	63024.5		IEEE =	63024.5		Match: YES
i=90000		NPAp =	70902.8		IEEE =	70902.8		Match: YES
i=100000		NPAp =	78781		IEEE =	78781		Match: YES

4. Sum of $x_1 = -0.893827$ with 999999
additions of $x_2 = 0.787827$

```
x1 : -0.893827
x2 : 0.787827
Z : 1000000
```

i= 2		NPAp =	-0.106		IEEE =	-0.106		Match: YES
i= 3		NPAp =	0.681827		IEEE =	0.681827		Match: YES
i= 4		NPAp =	1.46965		IEEE =	1.46965		Match: YES
i= 5		NPAp =	2.25748		IEEE =	2.25748		Match: YES
i= 6		NPAp =	3.04531		IEEE =	3.04531		Match: YES
i= 7		NPAp =	3.83314		IEEE =	3.83314		Match: YES
i= 8		NPAp =	4.62096		IEEE =	4.62096		Match: YES
i= 9		NPAp =	5.40879		IEEE =	5.40879		Match: YES
i= 10		NPAp =	6.19662		IEEE =	6.19662		Match: YES
i=100000		NPAp =	78781		IEEE =	78781		Match: YES
i=200000		NPAp =	157564		IEEE =	157564		Match: YES
i=300000		NPAp =	236346		IEEE =	236346		Match: YES
i=400000		NPAp =	315129		IEEE =	315129		Match: YES
i=500000		NPAp =	393912		IEEE =	393912		Match: YES
i=600000		NPAp =	472695		IEEE =	472695		Match: YES
i=700000		NPAp =	551477		IEEE =	551477		Match: YES
i=800000		NPAp =	630260		IEEE =	630260		Match: YES
i=900000		NPAp =	709043		IEEE =	709043		Match: YES
i=1000000		NPAp =	787825		IEEE =	787825		Match: YES

5. Sum of $x_1 = -8.93827e+10$ with 999
additions of $x_2 = 7.87827e+09$

```
x1 : -8.93827e+10
x2 : 7.87827e+09
Z : 1000
```

i= 2		NPAp =	-8.15044e+10		IEEE =	-8.15044e+10		Match: YES
i= 3		NPAp =	-7.36262e+10		IEEE =	-7.36262e+10		Match: YES
i= 4		NPAp =	-6.57479e+10		IEEE =	-6.57479e+10		Match: YES
i= 5		NPAp =	-5.78696e+10		IEEE =	-5.78696e+10		Match: YES
i= 6		NPAp =	-4.99914e+10		IEEE =	-4.99914e+10		Match: YES
i= 7		NPAp =	-4.21131e+10		IEEE =	-4.21131e+10		Match: YES
i= 8		NPAp =	-3.42348e+10		IEEE =	-3.42348e+10		Match: YES
i= 9		NPAp =	-2.63565e+10		IEEE =	-2.63565e+10		Match: YES
i= 10		NPAp =	-1.84783e+10		IEEE =	-1.84783e+10		Match: YES
i=100		NPAp =	6.90566e+11		IEEE =	6.90566e+11		Match: YES
i=200		NPAp =	1.47839e+12		IEEE =	1.47839e+12		Match: YES
i=300		NPAp =	2.26622e+12		IEEE =	2.26622e+12		Match: YES
i=400		NPAp =	3.05405e+12		IEEE =	3.05405e+12		Match: YES
i=500		NPAp =	3.84187e+12		IEEE =	3.84187e+12		Match: YES
i=600		NPAp =	4.6297e+12		IEEE =	4.6297e+12		Match: YES
i=700		NPAp =	5.41753e+12		IEEE =	5.41753e+12		Match: YES
i=800		NPAp =	6.20536e+12		IEEE =	6.20536e+12		Match: YES
i=900		NPAp =	6.99318e+12		IEEE =	6.99318e+12		Match: YES
i=1000		NPAp =	7.78101e+12		IEEE =	7.78101e+12		Match: YES

6. Sum of $x_1 = -8.93827e+10$ with 9999
additions of $x_2 = 7.87827e+09$

```
x1 : -8.93827e+10
x2 : 7.87827e+09
Z : 10000
```

i= 2		NPAp =	-8.15044e+10		IEEE =	-8.15044e+10		Match: YES
i= 3		NPAp =	-7.36262e+10		IEEE =	-7.36262e+10		Match: YES
i= 4		NPAp =	-6.57479e+10		IEEE =	-6.57479e+10		Match: YES
i= 5		NPAp =	-5.78696e+10		IEEE =	-5.78696e+10		Match: YES
i= 6		NPAp =	-4.99914e+10		IEEE =	-4.99914e+10		Match: YES
i= 7		NPAp =	-4.21131e+10		IEEE =	-4.21131e+10		Match: YES
i= 8		NPAp =	-3.42348e+10		IEEE =	-3.42348e+10		Match: YES
i= 9		NPAp =	-2.63565e+10		IEEE =	-2.63565e+10		Match: YES
i= 10		NPAp =	-1.84783e+10		IEEE =	-1.84783e+10		Match: YES
i=1000		NPAp =	7.78101e+12		IEEE =	7.78101e+12		Match: YES
i=2000		NPAp =	1.56593e+13		IEEE =	1.56593e+13		Match: YES
i=3000		NPAp =	2.35376e+13		IEEE =	2.35376e+13		Match: YES
i=4000		NPAp =	3.14158e+13		IEEE =	3.14158e+13		Match: YES
i=5000		NPAp =	3.92941e+13		IEEE =	3.92941e+13		Match: YES
i=6000		NPAp =	4.71724e+13		IEEE =	4.71724e+13		Match: YES
i=7000		NPAp =	5.50506e+13		IEEE =	5.50506e+13		Match: YES
i=8000		NPAp =	6.29289e+13		IEEE =	6.29289e+13		Match: YES
i=9000		NPAp =	7.08072e+13		IEEE =	7.08072e+13		Match: YES
i=10000		NPAp =	7.86855e+13		IEEE =	7.86855e+13		Match: YES

7. Sum of $x_1 = -8.93827e+10$ with 99999 additions of $x_2 = 7.87827e+09$

```
x1 : -8.93827e+10
x2 : 7.87827e+09
Z : 100000
```

i= 2		NPAp =	-8.15044e+10		IEEE =	-8.15044e+10		Match: YES
i= 3		NPAp =	-7.36262e+10		IEEE =	-7.36262e+10		Match: YES
i= 4		NPAp =	-6.57479e+10		IEEE =	-6.57479e+10		Match: YES
i= 5		NPAp =	-5.78696e+10		IEEE =	-5.78696e+10		Match: YES
i= 6		NPAp =	-4.99914e+10		IEEE =	-4.99914e+10		Match: YES
i= 7		NPAp =	-4.21131e+10		IEEE =	-4.21131e+10		Match: YES
i= 8		NPAp =	-3.42348e+10		IEEE =	-3.42348e+10		Match: YES
i= 9		NPAp =	-2.63565e+10		IEEE =	-2.63565e+10		Match: YES
i= 10		NPAp =	-1.84783e+10		IEEE =	-1.84783e+10		Match: YES
i=10000		NPAp =	7.86855e+13		IEEE =	7.86855e+13		Match: YES
i=20000		NPAp =	1.57468e+14		IEEE =	1.57468e+14		Match: YES
i=30000		NPAp =	2.36251e+14		IEEE =	2.36251e+14		Match: YES
i=40000		NPAp =	3.15034e+14		IEEE =	3.15034e+14		Match: YES
i=50000		NPAp =	3.93816e+14		IEEE =	3.93816e+14		Match: YES
i=60000		NPAp =	4.72599e+14		IEEE =	4.72599e+14		Match: YES
i=70000		NPAp =	5.51382e+14		IEEE =	5.51382e+14		Match: YES
i=80000		NPAp =	6.30164e+14		IEEE =	6.30164e+14		Match: YES
i=90000		NPAp =	7.08947e+14		IEEE =	7.08947e+14		Match: YES
i=100000		NPAp =	7.8773e+14		IEEE =	7.8773e+14		Match: YES

8. Sum of $x_1 = -8.93827e+10$ with 999999
additions of $x_2 = 7.87827e+09$

```
x1 : -8.93827e+10
x2 : 7.87827e+09
Z : 1000000
```

i= 2		NPAp =	-8.15044e+10		IEEE =	-8.15044e+10		Match: YES
i= 3		NPAp =	-7.36262e+10		IEEE =	-7.36262e+10		Match: YES
i= 4		NPAp =	-6.57479e+10		IEEE =	-6.57479e+10		Match: YES
i= 5		NPAp =	-5.78696e+10		IEEE =	-5.78696e+10		Match: YES
i= 6		NPAp =	-4.99914e+10		IEEE =	-4.99914e+10		Match: YES
i= 7		NPAp =	-4.21131e+10		IEEE =	-4.21131e+10		Match: YES
i= 8		NPAp =	-3.42348e+10		IEEE =	-3.42348e+10		Match: YES
i= 9		NPAp =	-2.63565e+10		IEEE =	-2.63565e+10		Match: YES
i= 10		NPAp =	-1.84783e+10		IEEE =	-1.84783e+10		Match: YES
i=100000		NPAp =	7.8773e+14		IEEE =	7.8773e+14		Match: YES
i=200000		NPAp =	1.57556e+15		IEEE =	1.57556e+15		Match: YES
i=300000		NPAp =	2.36338e+15		IEEE =	2.36338e+15		Match: YES
i=400000		NPAp =	3.15121e+15		IEEE =	3.15121e+15		Match: YES
i=500000		NPAp =	3.93904e+15		IEEE =	3.93904e+15		Match: YES
i=600000		NPAp =	4.72687e+15		IEEE =	4.72687e+15		Match: YES
i=700000		NPAp =	5.51469e+15		IEEE =	5.51469e+15		Match: YES
i=800000		NPAp =	6.30252e+15		IEEE =	6.30252e+15		Match: YES
i=900000		NPAp =	7.09035e+15		IEEE =	7.09035e+15		Match: YES

END